

Machine Technical Manual: Thermo-Compression Precision Die Attacher

Domain Intelligence & Anomaly Diagnostics Specification for Model DA-001

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Target Platform:	VectorAI Predictive Engine	Knowledge Role:	Primary Domain Source

SYNTHETIC PROTOTYPE TECHNICAL MANUAL — VECTORAI SYSTEM SPECIFICATION

This document is artificially generated specifically for the VectorAI predictive maintenance demonstration, knowledge base retrieval (RAG), formula-based RUL calculation, and automated diagnostics.

The specifications, thresholds, sensor values, service-life hours, maintenance intervals, and operating parameters are synthetic and must NOT be used for real-world physical industrial equipment operation or maintenance.

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1. Document Information & Governance

This technical document provides authoritative machine specifications for **Thermo-Compression Precision Die Attacher** (Identifier: DA-001, Manual ID: VAI-MAN-DA-001). It establishes the single source of truth for the VectorAI platform, defining telemetry sensor schemas, threshold limits, maintenance intervals, and deterministic degradation parameters.

Governance Principle: All diagnostic algorithms, RUL models, and threshold engines in VectorAI must strictly adhere to the limits and parameters defined herein. No machine learning regressions or stochastic models are permitted in the primary RUL calculation pipeline.

2. Machine Overview & Manufacturing Process

Process Stage: Die Attach & Bonding

High-speed, ultra-precision die bonding system designed to pick singulated semiconductor dies from diced wafer film frames and place them onto leadframes, organic substrates, or ceramic carriers with sub-micron positional accuracy using epoxy dispensing or eutectic bonding.

Detailed Process Flow:

The die attach cycle begins with wafer expansion and bottom-up ejector pin actuation. A high-speed pick-and-place robotic bond head fitted with a vacuum collet lifts the die. An epoxy dispensing unit applies programmable dot or cross patterns onto the leadframe pad. The heated substrate indexer stage maintains precise temperature while the bond head applies controlled compression force to achieve uniform bond line thickness.

Key Machine Subsystems:

- High-Speed Voice-Coil Pick-and-Place Bond Arm
- Substrate Indexer & Heated Workholder Stage
- Precision Augered Epoxy Dispense Pump
- Wafer Ejector Pin & Expansion Table
- Dual Optical Look-Up/Look-Down Vision Alignment Cameras
- Collet Vacuum & Programmable Positive Purge Manifold

3. Major Components & Degradation Characteristics

Component	Primary Function	Important Parameters	Degradation Indicators
Pick-and-Place Bond Head Arm	Transfers dies from diced wafer carrier to leadframe with precise Z-axis touchdown force and X-Y-theta orientation.	Arm linear traverse vibration (mm/s), voice-coil current (A), positioning repeatability (μm).	Traverse vibration spikes (>0.7 mm/s), bond position drift, linear encoder jitter.
Vacuum Collet / Pickup Tool	Secures delicate silicon die surface via suction during high acceleration (up to 10G) transit.	Collet vacuum pressure (kPa), tip wear/contamination, positive blow-off purge time (ms).	Vacuum leakage (>-55 kPa), die dropping, collet rubber tip deformation, silicone residue.
Epoxy Dispense System	Applies micro-liter volumes of thermally or electrically conductive adhesive to target bond pads.	Dispense fluid pressure (bar), syringe temperature (°C), needle Z-gap (μm), dot volume repeatability.	Dispense pressure creep (>5.5 bar), epoxy tailing, dispensing void formation, needle partial clogging.
Heated Substrate Indexer Stage	Heats leadframe/substrate to facilitate adhesive curing or eutectic bonding while indexing strip carriers.	Heater block temperature (°C), temperature uniformity across heater zone (± 2°C), clamp pressure.	Heater block temperature drift (>195°C), heater cartridge resistance imbalance, leadframe warping.
Wafer Ejector Pin Assembly	Pushes up through dicing tape under target die to delaminate die backside from adhesive tape.	Ejector pin travel height (μm), ascent velocity (mm/s), pin tip radius of curvature.	Die cracking, tape piercing, ejector guide bushing play, pin tip deformation.

4. Sensor Configuration & Telemetry Mapping

The following sensor schema represents the continuous telemetry parameters streamed by the machine controller into the VectorAI Edge Collector. Direction indicates whether increasing or decreasing values signify degradation.

Sensor ID	Sensor Name	Unit	Normal Range	Warning Range	Critical Range	Direction
vibration_arm	Arm Vibration	mm/s	0.1 – 0.4	0.4 – 0.7	0.7 – 2.0	HIGHER IS WORSE
pressure_vacuum	Collet Vacuum Pressure	kPa	-85.0 – -70.0	-70.0 – -55.0	-55.0 – 0.0	HIGHER IS WORSE
temperature_heater	Heater Block Temp	°C	140.0 – 180.0	180.0 – 195.0	195.0 – 260.0	HIGHER IS WORSE
bond_force	Placement Bond Force	N	4.5 – 6.5	6.5 – 8.0	8.0 – 15.0	HIGHER IS WORSE
dispense_pressure	Epoxy Dispense Pressure	bar	3.5 – 4.5	4.5 – 5.5	5.5 – 8.0	HIGHER IS WORSE
placement_offset	Placement Positional Offset	µm	0.0 – 5.0	5.0 – 10.0	10.0 – 25.0	HIGHER IS WORSE

5. Sensor Threshold Definitions & Operational Boundaries

Sensor / Metric	Normal Operational Band	Warning Threshold Band	Critical Fault Band
Arm Vibration vibration_arm	0.1 to 0.4 mm/s Smooth linear motion with well-damped voice-coil servo loop.	0.4 to 0.7 mm/s Linear guide rail dry friction or slight servo gain oscillation.	0.7 to 2.0 mm/s Severe mechanical guide binding or loose bond head mounting assembly.
Collet Vacuum Pressure pressure_vacuum	-85.0 to -70.0 kPa Firm suction seal holding silicon die without slippage.	-70.0 to -55.0 kPa Collet tip rubber wear, minor particulate leak, or tape residue on collet.	-55.0 to 0.0 kPa Vacuum seal compromised; high probability of dropped die during transfer.
Heater Block Temp temperature_heater	140.0 to 180.0 °C Target curing and wetting temperature profile for silver epoxy.	180.0 to 195.0 °C Thermal overshoot; risk of accelerated resin pre-curing on nozzle.	195.0 to 260.0 °C Severe thermal runaway risking substrate oxidation and epoxy degradation.
Placement Bond Force bond_force	4.5 to 6.5 N Optimal compression for uniform adhesive bond line thickness (BLT).	6.5 to 8.0 N Excessive force causing epoxy squeeze-out and potential die edge micro-cracking.	8.0 to 15.0 N Excessive compression force; high probability of silicon die fracture.
Epoxy Dispense Pressure dispense_pressure	3.5 to 4.5 bar Nominal fluid delivery through dispense needle.	4.5 to 5.5 bar Epoxy viscosity increase due to pot-life expiration or partial needle clogging.	5.5 to 8.0 bar Severe needle clogging or cured epoxy blockage in dispensing valve.
Placement Positional Offset placement_offset	0.0 to 5.0 µm Within high-yield precision placement envelope.	5.0 to 10.0 µm Optical calibration drift or thermal expansion offset in staging.	10.0 to 25.0 µm Gross misalignment risking bond pad shorts or wire bonding failure.

6. Standard Operating Conditions & Environmental Limits

Ambient Temperature:	21.0 - 23.0 °C (Cleanroom ISO Class 6)
Relative Humidity:	40% - 50% RH non-condensing
Normal Operating Temperature:	150.0 - 175.0 °C (Substrate Heater)
Operating Pressure:	0.50 - 0.60 MPa (CDA pneumatic supply)
Operating Speed / Velocity:	4,500 - 8,000 Units Per Hour (UPH)
Standard Cycle Time:	0.45 - 0.80 seconds per placed die
Production Schedule:	20 - 24 hours/day continuous
Recommended Operating Conditions:	Epoxy syringe must be thawed at room temperature for 60 min before loading; syringe pot-life < 8 hours.
Max Continuous Operation:	120 hours before required dispense calibration and collet replacement

7. Maintenance Schedule & Expected Service Life

Component	Interval (Hours)	Expected Life	Required Maintenance Action & Procedure
Pickup Collet & Vacuum Tip	150 hrs	350 hrs	Inspect rubber/ceramic tip for wear, clean with alcohol swab, or install replacement collet. <i>Procedure:</i> Use collet microscope gauge to verify zero tip chipping. Perform vacuum pull test on calibration die.
Epoxy Dispensing Valve & Needle	24 hrs	500 hrs	Flush fluidics, purge expired epoxy, clean needle tip or replace with new calibrated gauge needle. <i>Procedure:</i> Purge 10 test dots onto calibration sheet. Measure dot diameter repeatability with vision system (target $\pm 5\%$).
Bond Head Linear Guide & Voice Coil	600 hrs	5000 hrs	Clean linear rails, lubricate guide bearings with cleanroom synthetic grease, calibrate voice-coil response. <i>Procedure:</i> Perform automated friction compensation test in diagnostic menu. Verify smooth Z-travel.
Wafer Ejector Pin Block	400 hrs	2500 hrs	Inspect pin sharpness under microscope, check pin height synchronization, replace bent pins. <i>Procedure:</i> Verify all 4 pins extend to identical height within 5 μm using optical depth sensor.
Substrate Heater Block & Thermocouples	800 hrs	6000 hrs	Check thermocouple contact resistance, verify thermal profile uniformity across all indexing zones. <i>Procedure:</i> Place multi-channel thermal logger on heater surface; verify temperature variance $< \pm 2^\circ\text{C}$ across surface.

8. Degradation Indicators & Physics of Failure

Each parameter below reflects physical mechanical or electrical wear in machine subsystems. The VectorAI anomaly engine monitors these indicators to calculate Remaining Useful Life (RUL).

Parameter	Normal Baseline	Degraded Condition	Critical Limit	Underlying Physics / Mechanism
Arm Traverse Vibration	Vibration amplitude between 0.10 and 0.40 mm/s during rapid pick-and-place motions.	Vibration climbs into 0.40 - 0.70 mm/s due to linear guide bearing wear or servo damping loss.	Vibration exceeds 0.70 mm/s; mechanical resonance causes die position jitter during placement.	Indicates mechanical raceway pitting on linear slides or loosened mechanical fasteners on bond head carriage.
Collet Vacuum Pressure	Vacuum seal maintains deep negative pressure between -85 and -70 kPa.	Vacuum degrades into -70 to -55 kPa range as rubber tip hardens or collects silicone dust.	Vacuum drops worse than -55 kPa; suction insufficient to hold die during high-acceleration travel.	Indicates elastomer tip fatigue, particulate clogging in collet vacuum channel, or solenoid valve seal degradation.
Heater Block Temperature	Stable temperature holding at 140 - 180 °C.	Temperature drifts to 180 - 195 °C due to thermocouple contact oxidation or solid-state relay aging.	Temperature exceeds 195 °C; high risk of epoxy premature skinning and substrate thermal stress.	Reflects heating element degradation, loose thermocouple sensor mounting, or PID loop tuning deviation.
Placement Bond Force	Compression force stable at 4.5 - 6.5 N.	Force creeps into 6.5 - 8.0 N range as voice coil calibration drifts.	Force exceeds 8.0 N; high risk of micro-fracturing delicate ultra-thin silicon dies.	Indicates load cell amplifier drift, voice coil magnet thermal demagnetization, or mechanical guide binding.
Epoxy Dispense Pressure	Fluid delivery pressure steady at 3.5 - 4.5 bar.	Pressure climbs into 4.5 - 5.5 bar to push increasingly viscous epoxy.	Pressure exceeds 5.5 bar; fluidics nearing complete blockage.	Direct indicator of epoxy polymerization in the fluid line or partial dried resin clogging the dispense nozzle orifice.

9. Deterministic Formula-Based RUL Model Configuration

Base Useful Life (BUL): 5000 operating hours

Mathematical Formula: VectorAI computes RUL deterministically without neural networks or stochastic regression. For each parameter *i*:

- For HIGHER_IS_WORSE: $d_i = (currentValue - healthyLimit) / (criticalLimit - healthyLimit)$
- For LOWER_IS_WORSE: $d_i = (healthyLimit - currentValue) / (healthyLimit - criticalLimit)$
- Clamp: $d_i = \max(0.0, \min(1.0, d_i))$
- Overall Degradation: $D_{total} = \sum (w_i \times d_i)$ [where $\sum w_i = 1.00$]
- Remaining Useful Life: $RUL = \max(0, BaseUsefulLife \times (1.0 - D_{total}))$

Parameter	Sensor ID	Unit	Weight (w_i)	Healthy Limit	Critical Limit	Direction
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Arm Vibration	vibration_arm	mm/s	0.25	0.4	1.0	HIGHER IS WORSE
Collet Vacuum	pressure_vacuum	kPa	0.25	-70.0	-45.0	HIGHER IS WORSE
Heater Temperature	temperature_heater	°C	0.20	180.0	215.0	HIGHER IS WORSE
Placement Bond Force	bond_force	N	0.15	6.5	9.0	HIGHER IS WORSE
Dispense Pressure	dispense_pressure	bar	0.15	4.5	6.0	HIGHER IS WORSE
TOTAL WEIGHT SUM	-	-	1.00 (100%)	-	-	VERIFIED OK

10. Troubleshooting & Failure Symptoms Table

The following troubleshooting matrix provides rapid diagnostic mappings between observable anomalies and root causes.

Symptom & ID	Severity	Related Sensors	Possible Causes	Recommended Corrective Action
Collet Vacuum Drop / Die Dropping During Transfer SYM-DA-01	Critical	pressure_vacuum	• Pickup collet elastomer tip abrasion or tearing • Silicone/epoxy particulate accumulation inside collet vacuum lumen • Vacuum solenoid valve seat leakage	Replace pickup collet, clean vacuum filter line, verify holding vacuum reaches -80 kPa on calibration die.
Epoxy Dispensing Volume Shortage / Dot Tail-Off SYM-DA-02	High	dispense_pressur e	• Epoxy exceeding pot-life shelf time and undergoing room-temperature crosslinking • Partial curing of adhesive inside dispense needle cannula • Pneumatic pressure regulator valve diaphragm failure	Purge fluidics, replace dispense needle, thaw and install fresh epoxy syringe.
Pick-and-Place Bond Head Linear Traverse Jitter SYM-DA-03	Medium	vibration_arm, placement_offset	• Voice-coil linear motor magnetic track debris contamination • Linear guide rail dry friction due to grease degradation • Servo amplifier velocity feed-forward gain mistuning	Clean linear motor magnetic track, apply cleanroom grease to rails, re-run auto-tuning routine.
Substrate Stage Over-Temperature Condition SYM-DA-04	High	temperature_heat er	• Thermocouple probe mechanical contact separation from heater copper block • Solid-State Relay (SSR) short circuit failure in conductive mode • PID temperature controller loop saturation	Inspect and tighten thermocouple fastening screw, test SSR continuity, verify 160°C setpoint stability.
Die Micro-Cracking During Z-Placement Touchdown SYM-DA-05	Critical	bond_force, placement_offset	• Voice-coil touchdown impact force threshold set excessively high • Z-axis optical encoder scale missing counts during descent • Uneven epoxy thickness causing mechanical stress concentration	Perform Z-axis touchdown force calibration with load cell gauge (set to 5.0 N), inspect encoder scale.

11. Comprehensive Diagnostic Knowledge Base (Failure Scenarios)

This section contains 13 detailed synthetic failure scenarios used as the initial primary diagnostic knowledge base by the VectorAI Machine Agent. When an anomaly is detected, the agent queries this manual first before fallback to RAG.

Scenario 1: Collet Tip Wear Causing Die Release Failure (SCEN-DA-001)	Severity: HIGH
Telemetry Sensor Pattern:	pressure_vacuum degrades to -58 kPa; cycle time spikes
Possible Root Causes:	Rubber collet suction cup rim micro-tears Positive air purge blow-off solenoid response delayed Epoxy bleed adhesive contamination on collet perimeter
Recommended Corrective Action:	Replace pickup collet with new PEEK/rubber tool. Clean collet holder seat. Calibrate blow-off purge pulse duration.
Verification & Recovery Steps:	Test pick-and-place cycle on 20 dummy dies > Verify zero drop/stick errors
Scenario 2: Epoxy Dispense Needle Progressive Clogging (SCEN-DA-002)	Severity: HIGH
Telemetry Sensor Pattern:	dispense_pressure rises from 3.8 to 5.4 bar; dot size decreases by 30%
Possible Root Causes:	Conductive silver flakes settling and agglomerating at needle restriction Syringe temperature elevated above 28°C Epoxy pot-life exceeded 8 hours
Recommended Corrective Action:	Unload syringe, replace needle with new 25G precision cannula, load fresh batch of epoxy.
Verification & Recovery Steps:	Perform 5 weight test dispenses on precision balance > Verify dot weight within ± 3% target
Scenario 3: Bond Arm Voice Coil Vibration Resonance (SCEN-DA-003)	Severity: HIGH
Telemetry Sensor Pattern:	vibration_arm climbs to 0.78 mm/s; placement_offset reaches 8.5 μm
Possible Root Causes:	Bond head mounting screws loose due to repetitive high-G cycling Linear optical encoder tape dirt smudge Voice-coil flexible ribbon cable fatigue
Recommended Corrective Action:	Torque all bond head mounting bolts to 1.2 Nm. Clean optical encoder scale with anhydrous ethanol.
Verification & Recovery Steps:	Execute vibration frequency FFT sweep > Verify arm vibration returns to < 0.35 mm/s
Scenario 4: Substrate Stage Thermocouple Detachment (SCEN-DA-004)	Severity: HIGH
Telemetry Sensor Pattern:	temperature_heater reads erratic swings (130°C to 210°C in seconds)
Possible Root Causes:	Thermocouple clamp screw stripped Thermocouple lead wire shield frayed Heater cartridge internal resistance imbalance
Recommended Corrective Action:	Replace thermocouple assembly, verify thermal paste application, verify stable reading at 165°C.
Verification & Recovery Steps:	Log temperature for 15 minutes > Verify temperature ripple is < ± 1.0°C

Scenario 5: Wafer Ejector Pin Misalignment and Silicon Chipping (SCEN-DA-005)	Severity: CRITICAL
Telemetry Sensor Pattern:	placement_offset drifts; pressure_vacuum fluctuates during pick stroke
Possible Root Causes:	Central ejector pin tip bent from tape puncture collision Wafer expander ring tension relaxed Ejector Z-axis motor zero-position lost
Recommended Corrective Action:	Replace 4-pin ejector set. Re-teach wafer table ejector X-Y-Z coordinates with alignment fixture.
Verification & Recovery Steps:	Perform dry ejection test under camera inspection > Verify concentricity with collet
Scenario 6: Die Tilt & Uneven Epoxy Bond Line Thickness (SCEN-DA-006)	Severity: MEDIUM
Telemetry Sensor Pattern:	bond_force creeps to 7.6 N; placement_offset at 6.8 μm
Possible Root Causes:	Collet shank angle misaligned relative to substrate plane Heater stage clamping plate warped from thermal cycling Asymmetrical epoxy dot dispense volume
Recommended Corrective Action:	Perform optical planarity calibration on bond head. Inspect leadframe clamping insert flatness.
Verification & Recovery Steps:	Inspect bond line thickness on cross-section sample (target $25 \pm 3 \mu\text{m}$) > Verify die tilt < 5 μm
Scenario 7: Vision Alignment Optical Lens Fogging (SCEN-DA-007)	Severity: HIGH
Telemetry Sensor Pattern:	Placement offset increases to 11.2 μm ; camera lighting intensity alert
Possible Root Causes:	Epoxy volatile solvent vapor condensing on bottom look-down camera lens LED strobe illuminator brightness decay Focal height shift due to thermal expansion
Recommended Corrective Action:	Clean camera optical windows with lens fluid. Re-calibrate camera pixel-to-micron scaling factor.
Verification & Recovery Steps:	Run dot-grid calibration target > Verify vision repeatability < 0.8 μm
Scenario 8: Leadframe Indexing Stepper Motor Jam (SCEN-DA-008)	Severity: MEDIUM
Telemetry Sensor Pattern:	Cycle time alert; vibration_arm at 0.55 mm/s; line jam interlock
Possible Root Causes:	Leadframe strip bent or guide track width set too narrow Indexer feed claw wear or mechanical backlash Epoxy stringing onto indexing track guide
Recommended Corrective Action:	Clear leadframe jam, clean track rails with IPA, adjust track width with thickness shim.
Verification & Recovery Steps:	Index 50 leadframe strips automatically > Verify index pitch repeatability < 10 μm
Scenario 9: Dispense Syringe Temperature Controller Drift (SCEN-DA-009)	Severity: MEDIUM
Telemetry Sensor Pattern:	dispense_pressure creeping up; dot shape changing from round to oval
Possible Root Causes:	Peltier cooling/heating jacket fan blocked Syringe jacket RTD sensor loose Ambient cleanroom draft disrupting thermal balance
Recommended Corrective Action:	Clean cooling fan filters. Reseat RTD probe inside syringe thermal block.
Verification & Recovery Steps:	Verify syringe maintains $25.0 \pm 0.5^\circ\text{C}$ > Measure dispense consistency

Scenario 10: Load Cell Zero-Offset Saturation (SCEN-DA-010)	Severity: CRITICAL
Telemetry Sensor Pattern:	bond_force registers false baseline offset (+2.5 N at rest)
Possible Root Causes:	Bond head load cell strain gauge pre-stress Signal amplifier grounding loop noise Cable flex strain on sensor lead wires
Recommended Corrective Action:	Perform zero tare calibration in service utility. Check sensor cable strain relief.
Verification & Recovery Steps:	Verify 0.0 N reading with head unloaded > Test 5.0 N precision weight block
Scenario 11: Wafer Stage Vacuum Chuck Tape Slippage (SCEN-DA-011)	Severity: MEDIUM
Telemetry Sensor Pattern:	Ejector pin pickup failure rate increases; vision alignment offsets jump
Possible Root Causes:	Wafer stage vacuum level decayed below -60 kPa Porosity of stage ceramic disc blocked by adhesive residue Tape frame clamping ring latch loose
Recommended Corrective Action:	Clean ceramic wafer chuck with acetone wipe. Replace vacuum line seal.
Verification & Recovery Steps:	Verify wafer ring clamp lock tension > Verify -80 kPa vacuum under wafer frame
Scenario 12: Positive Blow-Off Air Pressure Regulator Leak (SCEN-DA-012)	Severity: MEDIUM
Telemetry Sensor Pattern:	pressure_vacuum fails to recover quickly after blow-off pulse
Possible Root Causes:	Blow-off solenoid valve plunger seal sticking Pneumatic pressure regulator diaphragm pinhole Debris in exhaust muffler
Recommended Corrective Action:	Replace blow-off pneumatic solenoid valve. Clean manifold exhaust port.
Verification & Recovery Steps:	Measure vacuum release transition time (< 20 ms) > Verify complete die release
Scenario 13: Bond Arm Theta Rotation Motor Backlash (SCEN-DA-013)	Severity: HIGH
Telemetry Sensor Pattern:	placement_offset theta component drifts ± 0.5 degrees
Possible Root Causes:	Direct-drive theta motor bearing play Theta rotary encoder disk contamination Collet holder clamp set screw loose
Recommended Corrective Action:	Inspect and tighten collet holder set screw. Execute theta homing and alignment calibration.
Verification & Recovery Steps:	Measure rotational alignment accuracy on glass calibration die (< 0.1 deg)

12. Standard Operating & Maintenance Procedures (SOP)

SOP-01: Shift Start Operational Inspection & Tool Touch-Off

- 1. Verify Clean Dry Air (CDA) supply pressure is within 0.55 - 0.65 MPa and dew point is < -40°C.
- 2. Ensure cleanroom ambient temperature is stabilized within specified operating window (20 - 24°C).
- 3. Run automated sensor zero-calibration and optical target homing sequence.
- 4. Process 2 calibration test pieces; inspect output dimensions and telemetry stability on VectorAI Dashboard.

SOP-02: Tool Replacement & Calibration Protocol

- 1. Lock out and tag out (LOTO) machine controller before entering mechanical enclosure.
- 2. Use calibrated preset torque wrench to fasten tool fasteners to exact manufacturer specification.
- 3. Execute optical non-contact tool alignment and measure runout / contact resistance.
- 4. Log maintenance record in VectorAI with work order ID, technician ID, and timestamp.

SOP-03: Anomaly Escalation Protocol

When a sensor crosses from Warning into Critical threshold band:

- 1. VectorAI generates automated Severity: High/Critical Incident.
- 2. System automatically references Section 11 Diagnostic Knowledge for recommended immediate actions.
- 3. If Remaining Useful Life (RUL) falls below 48 hours, dynamic production rerouting is triggered to alternate healthy lines.

13. Document Revision History

Revision	Release Date	Author / Organization	Description of Changes
v1.0	2026-08-26	VectorAI Knowledge Engineering	Initial synthetic technical specification and diagnostic manual release.
v0.9-RC	2026-08-15	VectorAI Systems Lab	Draft sensor thresholds, RUL weight configuration, and failure scenarios.

14. VectorAI RAG Integration & Knowledge Indexing

This document is formatted with distinct semantic headings, tables, and structured diagnostic scenarios to facilitate high-precision chunking and vector embedding in the VectorAI RAG pipeline.

- **Chunk Association:** Every indexed chunk retains metadata: machineId, manualId, section.
- **Scoped Retrieval:** Diagnostics for machine DA-001 prioritize chunks where machineId == 'DA-001'.